Higher-order Matching via Intersection Type Inhabitation in Bounded Dimension

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Higher-order matching (HOM)

- Problem:
  - Given a pair of STLC terms $l = r$
  - Find a substitution S in STLC such that $S(l) =_{\beta\eta} r$
- Example:
  $f(X(\lambda y_1 y_2.y_1)ab)(X(\lambda y_1 y_2.y_2)ab) = f(fab)(fbb)$

  Solution: $[X := \lambda x_1 x_2 x_3.f(x_1 x_2 x_3)x_3]$

Alternative presentation

- HOM is equivalent to dual interpolation (DI, Padovani)
- A higher-order matching instance can be presented as a pair of systems of shape

$$X\, u_i^1 \, \ldots \, u_i^n =_{\beta\eta} r_i \qquad (i \in \{1, \ldots, m\}) \tag{1}$$

or shape

$$X\, u_i^1 \, \ldots \, u_i^n \neq_{\beta\eta} r_i \qquad (i \in \{1, \ldots, m\}) \tag{2}$$

- Example:
$\{X(\lambda y_1 y_2 . y_1) ab = fab,$
$X(\lambda y_1 y_2 . y_2) ab = fbb\}$

$\{X(\lambda y_1 y_2 . y_1) ab \neq a\}$

Solution: $[X := \lambda x_1 x_2 x_3 . f(x_1 x_2 x_3) x_3]$

Motivation

- Higher-order matching is applicable in
  - theorem proving
  - program transformation and
  - higher-order rewriting systems
- Decidability proof by Colin Stirling is too complicated
- Consequently $\Rightarrow$ we do not understand the nature of the problem
- Goal: make a clear decidability argument
- Clarity criterion: so that it can be formalised

Main Idea

- Use intersection types to semantically describe solutions of DI
- Employ inhabitation to find a solution
- Obstacle: inhabitation in ITD is undecidable
- Observation:
  - known undecidability proofs operate on vectors of target types
  - these vectors represent contents of a tape of a Turing Machine
- Idea: bound the culprit of undecidability, i.e. vector dimension

Format to express bounded dimension

- Introduce **System R**
- Typing judgements return **vectors of types**
- Use **relations** to control information flow
- Define **dimension** as vector length

Intersection Types

Definition

$$A ::= a \mid \sigma \to A$$

$$\sigma ::= \{A_1, \ldots, A_n\}$$

- Types built from atoms and arrow constructors
- Sets represent intersections

Type Vectors

- Work with vectors: $\bar{A} = (A_1, \ldots, A_n)$
- Dimension $=$ number of components
- Enables parallel typing

Relations

- Relations map between vector coordinates
- Control information flow

Example

$$R((a, b, c)) = (\{a, b\}, \emptyset, \{a, c\})$$

Typing Rules (System R)

- Judgements: $\Gamma \vdash_k t : \bar{A}$

Rules

$$\frac{\bar{A} \mathrel{\bar{\in}} \bar{\sigma} \qquad \dim(\bar{A}) \leq k}{\Gamma, x : \bar{\sigma} \vdash_k x : \bar{A}} \text{(Ax)}$$

$$\frac{\Gamma, x : \bar{\sigma} \vdash_k t : \bar{A}}{\Gamma \vdash_k \lambda x.t : \bar{\sigma} \Rightarrow \bar{A}} (\Rightarrow\text{I})$$

$$\frac{\Gamma \vdash_k t : R(\bar{A}) \Rightarrow \bar{B} \qquad \Delta \vdash_k u : \bar{A}}{\Gamma \cup R(\Delta) \vdash_k t\ u : \bar{B}} (\Rightarrow\text{E})$$

- k is the dimension bound
- vectors are of dimension at most k, see the (Ax) rule

Results for System R

- Subject reduction holds
- Subformula property limits search space
- Strong normalisation inherited
- Inhabitation decidable in:
  - 2-EXPTIME (bounded dimension)
  - EXPTIME (fixed dimension)
- Lies between:
  - multiset system (EXPSPACE)
  - set system (undecidable)

Characteristic Typings

- Consider rhs r of an interpolation equation (β-normal η-long term)
- We have $\Phi \vdash_{\mathsf{STLC}} r : \varphi$
- Devise
  - a refining environment Γ and
  - intersection type A

  such that

  for each η-long t
  $\Gamma \vdash_k t : A$ for some k iff $t \twoheadrightarrow_\beta r$
- Written $\Gamma \prec \Phi$ and $A \prec \varphi$

Characteristic Typings – Idea

- Idea: associate a unique type atom with each subexpression of r
- A type expresses:

Property

A term s of type

$$\sigma_1 \to \cdots \to \sigma_n \to A$$

when applied to arguments of types $\sigma_1, \ldots, \sigma_n$ reduces to A.

Characteristic Typings – Example

- Instance: $X(\lambda y_1 y_2.y_1)ab = fab$
- Solution: $[X := \lambda x_1 x_2 x_3.f(x_1 x_2 x_3)x_3]$
- Type atoms: a, b, fab
- Typing: $\emptyset \vdash (\lambda x_1 x_2 x_3.f(x_1 x_2 x_3)x_3)\ \ (\lambda y_1 y_2.y_1)\ \ a\ \ b : fab$
- More detailed:

$$\begin{aligned}\emptyset \vdash (\lambda x_1 &: (\{\{a\} \to \emptyset \to a\}).\\ x_2 &: (\{a\}).\\ x_3 &: (\{b\}).\\ &\quad f(x_1 x_2 x_3)x_3)\ \ (\lambda y_1 : (\{a\}).y_2 : (\emptyset).y_1)\ \ a\ \ b : (fab)\end{aligned}$$

- Also characteristic typing for: $X(\lambda y_1 y_2.y_1)ab \neq a$

Characteristic Typings – Another Example

- Instance: $X(\lambda y_1 y_2.y_2)ab = fbb$
- Solution: $[X := \lambda x_1 x_2 x_3.f(x_1 x_2 x_3)x_3]$
- Type atoms: b, fbb
- Typing: $\emptyset \vdash (\lambda x_1 x_2 x_3.f(x_1 x_2 x_3)x_3)\ \ (\lambda y_1 y_2.y_2)\ \ a\ \ b : fab$
- More detailed:

$$
\begin{aligned}
\emptyset \vdash (\lambda x_1 &: (\{\emptyset \to \{b\} \to b\}). \\
x_2 &: (\{a\}). \\
x_3 &: (\{b\}). \\
&\quad f(x_1 x_2 x_3)x_3)\ \ (\lambda y_1 : (\emptyset).y_2 : (\{b\}).y_2)\ \ a\ \ b : (fbb)
\end{aligned}
$$

Characteristic Typings

Lemma (equality characterisation)

For each β-normal η-long r with $\Phi \vdash_{\mathsf{STLC}} r : \varphi$, there exist $\Gamma \prec \Phi$ and $A \prec \varphi$ such that:

1. $\Gamma \vdash r : A$ in system $\mathbf{R}$
2. for any η-long t with $\Phi \vdash_{\mathsf{STLC}} t : \varphi$, if $\Gamma \vdash t : A$ then $t \twoheadrightarrow_\beta r$.

Lemma (disequality characterisation)

For each β-normal η-long r with $\Phi \vdash_{\mathsf{STLC}} r : \varphi$, there exist $\Gamma \prec \Phi$ and $A \prec \varphi$ such that for any η-long t with $\Phi \vdash_{\mathsf{STLC}} t : \varphi$ and $\Gamma \vdash t : A$ we have $t \not\twoheadrightarrow_\beta r$.

Reduction to Inhabitation

- We focus only on $\beta\eta$-equations
- Suppose $s = \lambda x_1 \ldots x_n.t$ is a solution to $\{X u_i^1 \ldots u_i^n = r_i\}_{i \in I}$
- Then $s\, u_i^1 \ \ldots \ u_i^n \twoheadrightarrow_\beta r_i$ for every i.
- There is a dimension bound k such that

$$\Gamma_{[i]} \vdash_k (\lambda x_1 \ldots x_n.t)\, u_i^1 \ \ldots \ u_i^n : \bar{A}_{[i]} \qquad (i \in \{1, \ldots, m\}).$$

- By inversion of rule ($\Rightarrow$E), there are some relations R_i^j and vectors $\bar{B}_i^j$ with

$$\Gamma_{[i]} \vdash_k \lambda x_1 \ldots x_n.t : R_i^1(\bar{B}_i^1) \Rightarrow \cdots \Rightarrow R_i^n(\bar{B}_i^n) \Rightarrow \bar{A}_{[i]},$$
$$\Gamma_{[i]} \vdash_k u_i^j : \bar{B}_i^j.$$

- Stack m derivations into vectors of dimension at most mk yields $R^1, \ldots, R^n$ and $\bar{B}^1, \ldots, \bar{B}^n$ such that

$$\Gamma, x_1 : \bar{B}^1, \ldots, x_n : \bar{B}^n \vdash_{mk} t : R^1(\bar{B}^1) \Rightarrow \cdots \Rightarrow R^n(\bar{B}^n) \Rightarrow \bar{A}$$

- t is an inhabitant of dimension mk, with environment and target type computable from $r_1, \ldots, r_m$ and the arguments u_i^j.

Reduction to Inhabitation

Proposition

The higher-order matching instance $\{X u_i^1 \dots u_i^n = r_i\}_{i \in I}$ has a solution

iff

the type $R^1(\bar{B}^1) \Rightarrow \dots \Rightarrow R^n(\bar{B}^n) \Rightarrow \bar{A}$ is inhabited in the environment $\Gamma, x_1 : \bar{B}^1, \dots, x_n : \bar{B}^n$ for some dimensional bound.

Dimension Bound for 4th Order

- Bounded subject expansion
- Padovani-style saturation
- Krivine-machine states

Stirling's Example

$$
\begin{aligned}
u_1 &= \lambda y_1^{o}.\, \lambda y_2^{o\to o\to o}.\, y_2\, y_1\, y_1 \\
u_2 &= \lambda y_3^{o\to(o\to o\to o)\to o}.\, y_3\, \Big(y_3\, a\, (\lambda w_1^{o}.\, \lambda w_2^{o}.\, w_1)\Big)\, (\lambda v_1^{o}.\, \lambda v_2^{o}.\, g\, v_2) \\
s &= \lambda x_1^{o\to(o\to o\to o)\to o}.\, \lambda x_2^{(o\to(o\to o\to o)\to o)\to o}. \\
&\qquad x_2\, \Big(\lambda z_1^{o}.\, \lambda z_2^{o\to o\to o}.\, x_1\, z_1\, (\lambda s_1^{o}.\, \lambda s_2^{o}.\, x_1\, b\, (\lambda u_1^{o}.\, \lambda u_2^{o}. \\
&\qquad\qquad z_2\, s_1\, s_2))\Big)
\end{aligned}
$$

$$
\begin{aligned}
s' &= \lambda x_1^{o\to(o\to o\to o)\to o}.\, \lambda x_2^{(o\to(o\to o\to o)\to o)\to o}. \\
&\qquad x_2\, \Big(\lambda z_1^{o}.\, \lambda z_2^{o\to o\to o}.\, x_1\, z_1\, (\lambda s_1^{o}.\, \lambda s_2^{o}.\, x_1\, b\, (\lambda u_1^{o}.\, \lambda u_2^{o}. \\
&\qquad\qquad z_2\, \underline{(x_1 z_1 (\lambda s_1 s_2. s_1))}\, s_2))\Big)
\end{aligned}
$$

Reduction of an overarching binding of s_1 over x_1.

Stirling's Example

$$
\begin{aligned}
u_1 = {} & \lambda y_1 : (\{a\}, \quad \emptyset, \quad \{a\}, \quad \emptyset). \\
& \lambda y_2 : (\{\{a\} \to \emptyset \to a\}, \quad \{\emptyset \to \emptyset \to a\}, \quad \{\emptyset \to a \to ga\}, \quad \{\emptyset \to \emptyset \to ga\}). \\
& \qquad y_2\, y_1\, y_1 : (a, a, ga, ga)
\end{aligned}
$$

$$
\begin{aligned}
u_2 = {} & \lambda y_3 : (\{\{a\} \to \{\{a\} \to \emptyset \to a\} \to a\}, \quad \{a\} \to \{\emptyset \to \{a\} \to ga\} \to ga\}). \\
& \qquad y_3\, \Big(y_3\, a\, (\lambda w_1 : \{a\}. \lambda w_2 : \emptyset. w_1)\Big)\, (\lambda v_1 : \emptyset. \lambda v_2 : \{a\}. g\, v_2) : (ga)
\end{aligned}
$$

$$
\begin{aligned}
s = {} & \lambda x_1 : (\{\{a\} \to \{\{a\} \to \emptyset \to a\} \to a, \\
& \qquad \emptyset \to \{\emptyset \to \emptyset \to a\} \to a, \\
& \qquad \{a\} \to \{\emptyset \to \{a\} \to ga\} \to ga, \\
& \qquad \emptyset \to \{\emptyset \to \emptyset \to ga\} \to ga\}). \\
& \lambda x_2 : (\{\{\{a\} \to \{\{a\} \to \emptyset \to a\} \to a, \ \{a\} \to \{\emptyset \to \{a\} \to ga\} \to ga\} \to ga\}). \\
& \qquad x_2\, \Big(\lambda z_1 : (\{a\}, \{a\}). \lambda z_2 : (\{\{a\} \to \emptyset \to a\}, \{\emptyset \to \{a\} \to ga\}). \\
& \qquad\qquad x_1\, z_1\, \big(\lambda s_1 : (\{a\}, \emptyset). \lambda s_2 : (\emptyset, \{a\}). \\
& \qquad\qquad\qquad x_1\, b\, (\lambda u_1 : (\emptyset, \emptyset). \lambda u_2 : (\emptyset, \emptyset). z_2\, s_1\, s_2)\big)\Big) : (ga)
\end{aligned}
$$

$x_1 z_1 (\lambda s_1 s_2 . s_1)$ has type (a)

Conclusion

- Alternative approach to decidability of the HOM problem
- Intersection types in format of System R
- Characteristic typings provide valuable reduction invariants
- Dimension restriction as a key to solution search
- Bound on dimension is the main technical difficulty